FDS

Fundamentals of Distributed systems

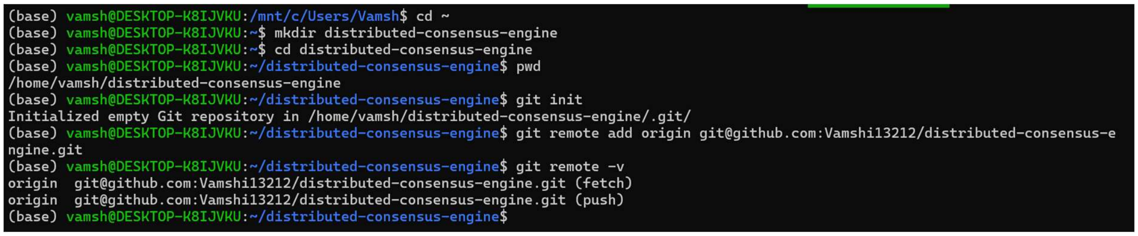
ASSIGNMENT – Q1

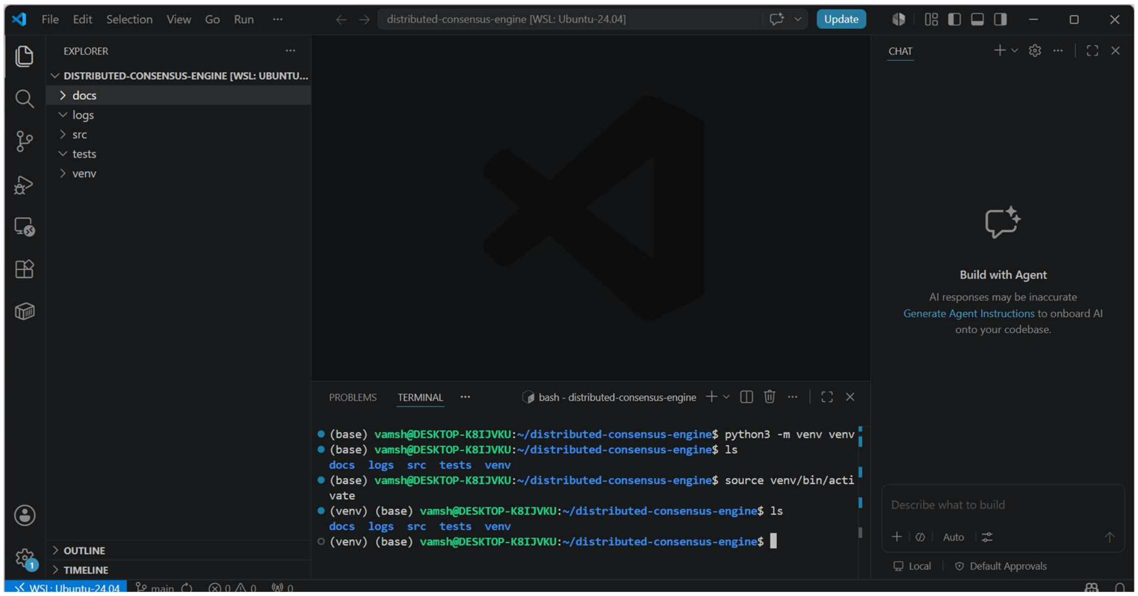
Resilient State Machine Replication: Paxos & PBFT

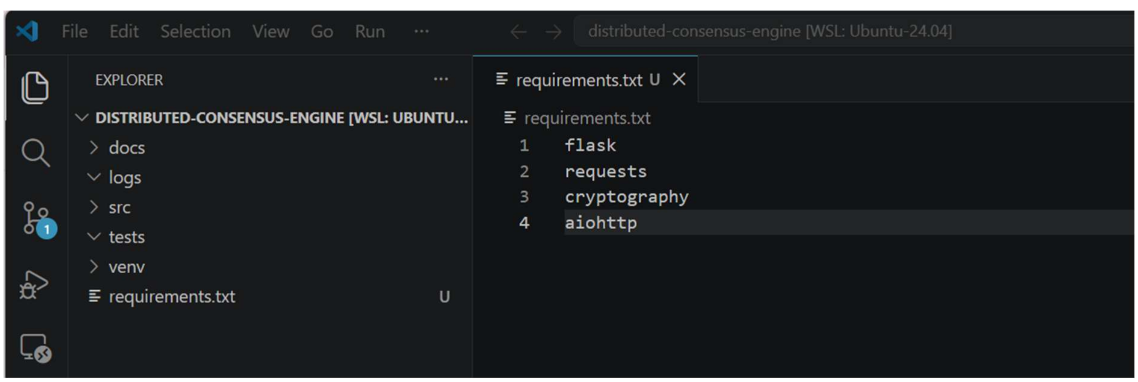
SUBMITTED BY

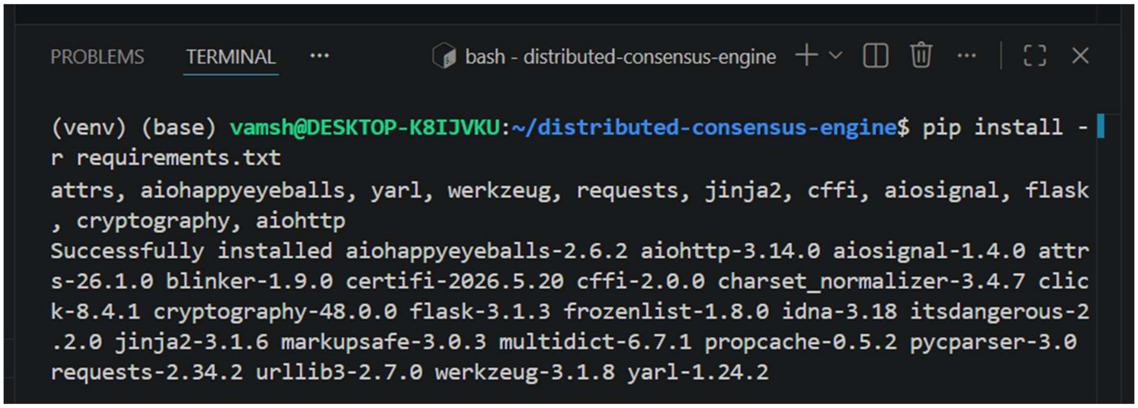
NAME Vamshikumar Bejjarapu	ROLL NUMBER G25AI1066
GitHub Repository	https://github.com/Vamshi13212/distributed-consensus-engine
Video Link	https://drive.google.com/file/d/12-pTDUUFmvT2rizmV5kLaYbk394e4FGa/view?usp=sharing

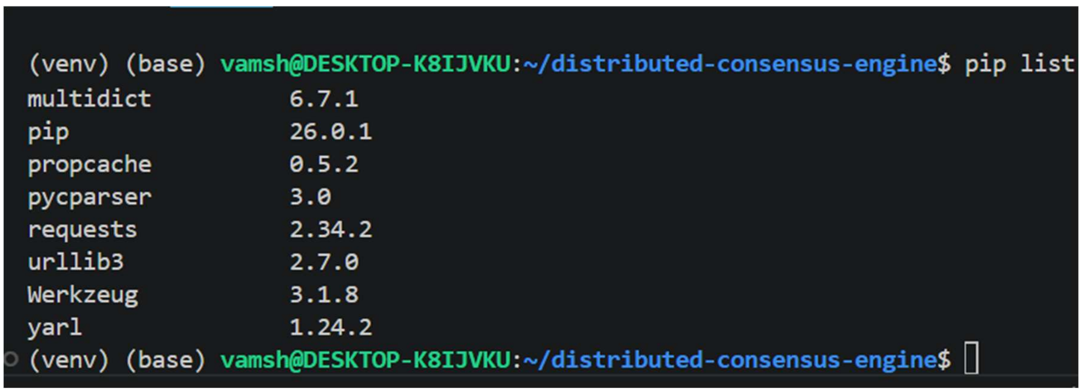
The below screenshot to show that repository was successfully initialized and linked to GitHub. The project directory structure was prepared for subsequent development phases


(base) vamsh@DESKTOP-K8IJVKU:/mnt/c/Users/Vamsh$ cd ~
(base) vamsh@DESKTOP-K8IJVKU:~$ mkdir distributed-consensus-engine
(base) vamsh@DESKTOP-K8IJVKU:~$ cd distributed-consensus-engine
(base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ pwd
/home/vamsh/distributed-consensus-engine
(base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ git init
Initialized empty Git repository in /home/vamsh/distributed-consensus-engine/.git/
(base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ git remote add origin git@github.com:Vamshi13212/distributed-consensus-e
ngine.git
(base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ git remote -v
origin git@github.com:Vamshi13212/distributed-consensus-engine.git (fetch)
origin git@github.com:Vamshi13212/distributed-consensus-engine.git (push)
(base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$

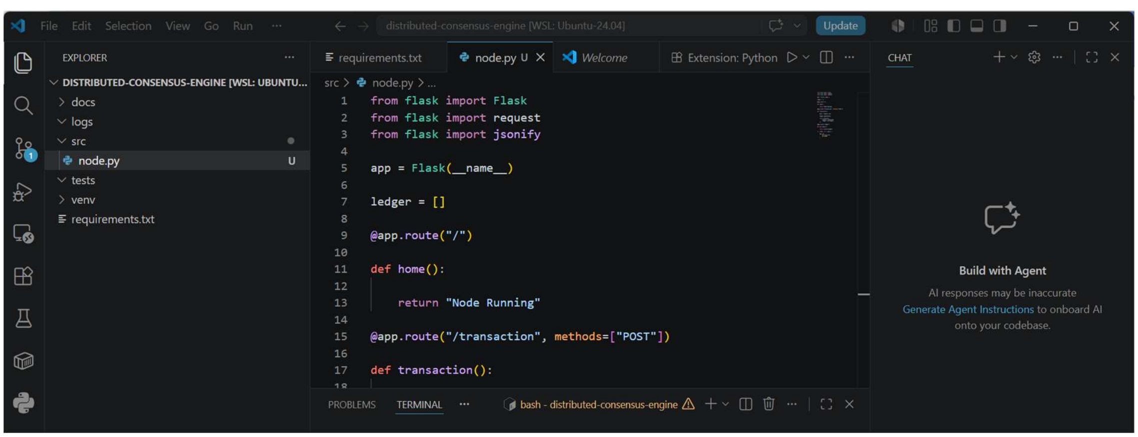

File Edit Selection View Go Run ...
distributed-consensus-engine [WSL: Ubuntu-24.04]
Update
EXPLORER
DISTRIBUTED-CONSENSUS-ENGINE [WSL: UBUNTU...
docs
logs
src
tests
venv
OUTLINE
TIMELINE
CHAT
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Describe what to build
Auto
Local
Default Approvals
PROBLEMS TERMINAL
bash - distributed-consensus-engine
(base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ python3 -m venv venv
(base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ ls
docs logs src tests venv
(base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ source venv/bin/acti
vate
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ ls
docs logs src tests venv
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$
WSL: Ubuntu-24.04
main


File Edit Selection View Go Run ...
distributed-consensus-engine [WSL: Ubuntu-24.04]
EXPLORER
DISTRIBUTED-CONSENSUS-ENGINE [WSL: UBUNTU...
docs
logs
src
tests
venv
requirements.txt U
requirements.txt U
requirements.txt
1 flask
2 requests
3 cryptography
4 aiohttp



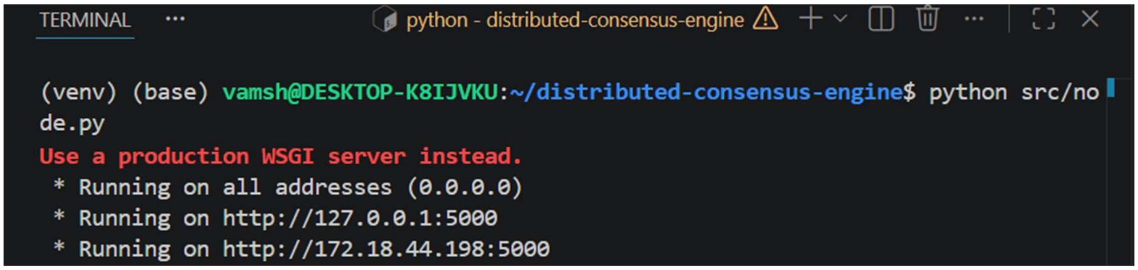


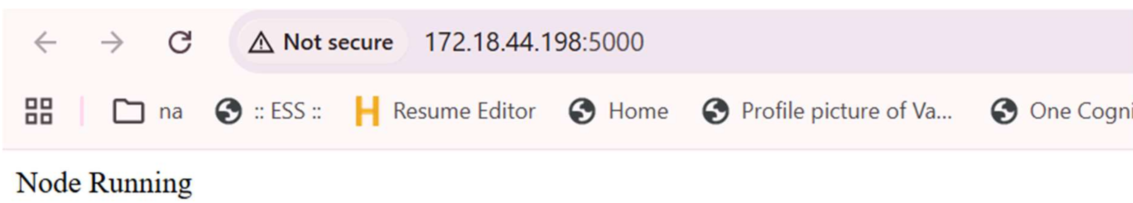
Create First Node



Run First Node

This figure shows the installation of required Python libraries, including Flask, Requests, Cryptography, and AIOHTTP. The screenshots also demonstrate the implementation of the first distributed node using Flask and the successful execution of the node service on port 5000.



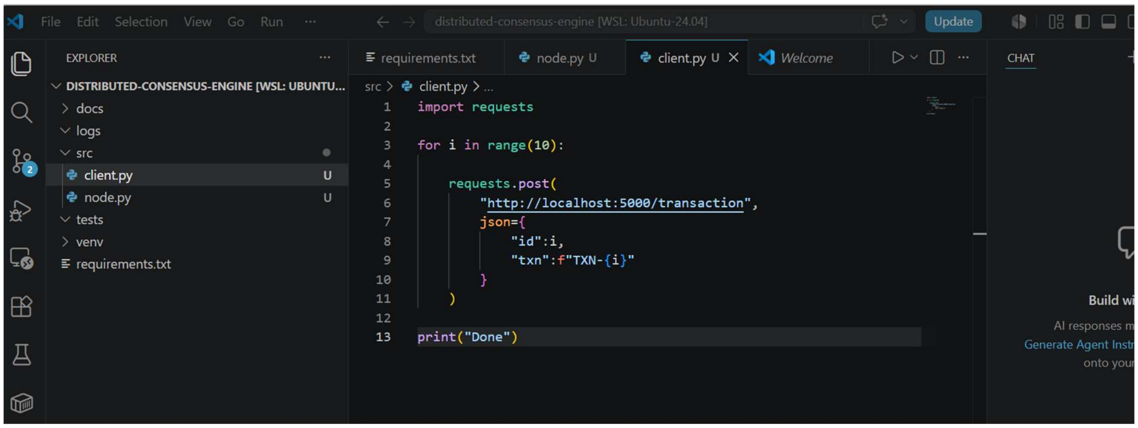


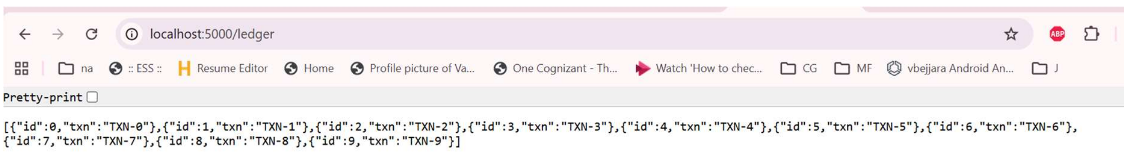
Create Client

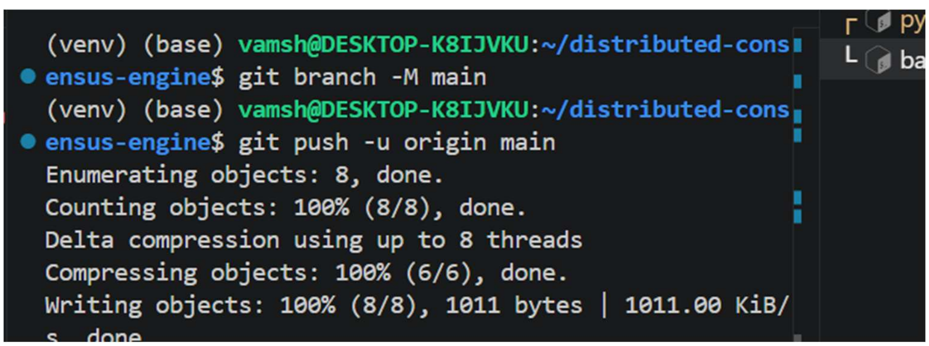
his figure presents the development of the client application responsible for generating and submitting transactions to the distributed node. The node service was executed and verified through browser access, while the client generated multiple transaction requests.

Observation: The client successfully communicated with the server using HTTP requests.

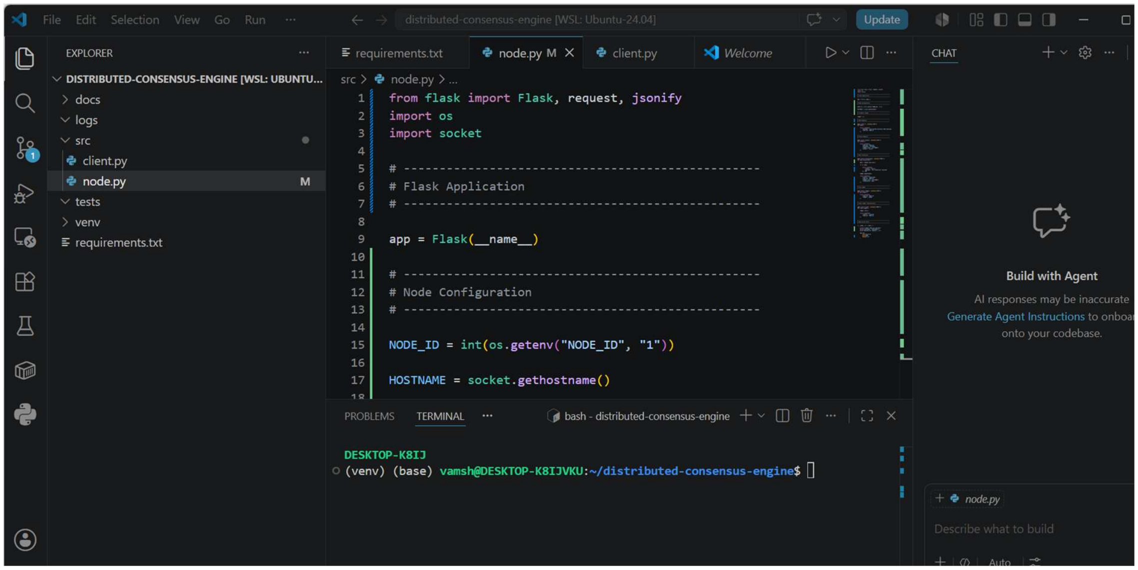
Outcome: Basic client-server communication was validated, confirming the correctness of the REST API implementation and transaction handling logic.

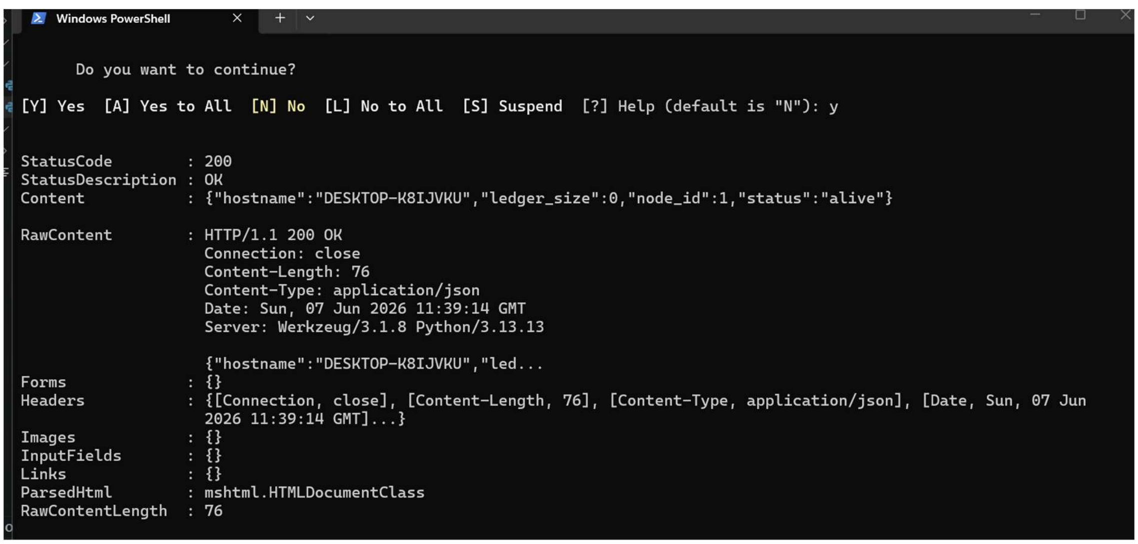




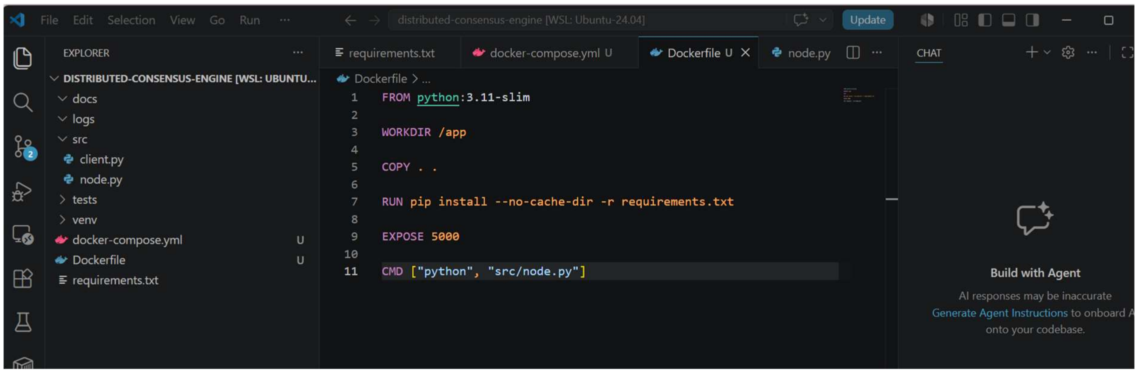


B**efore Leader Election and Paxos**. We first make sure all 5 nodes can communicate.

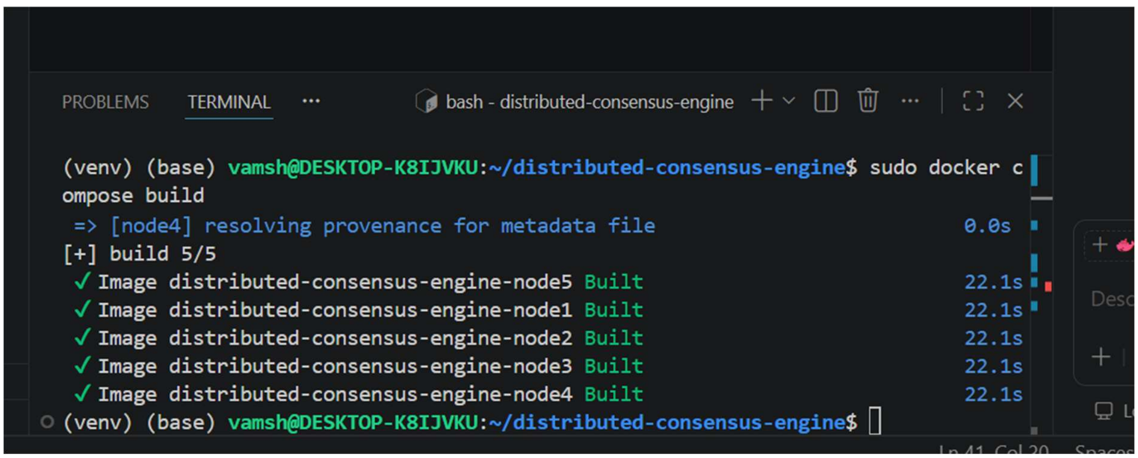

File Edit Selection View Go Run ···
distributed-consensus-engine [WSL: Ubuntu-24.04]
Update
EXPLORER
DISTRIBUTED-CONSENSUS-ENGINE [WSL: UBUNTU...
docs
logs
src
client.py
node.py M
tests
venv
requirements.txt
requirements.txt
node.py M
client.py
Welcome
src > node.py > ...
from flask import Flask, request, jsonify
import os
import socket
# ---
# Flask Application
# ---
app = Flask(__name__)
# ---
# Node Configuration
# ---
NODE_ID = int(os.getenv("NODE_ID", "1"))
HOSTNAME = socket.gethostname()
PROBLEMS TERMINAL ···
bash - distributed-consensus-engine
DESKTOP-K8IJ
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$
CHAT
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboar
onto your codebase.
node.py
Describe what to build
Auto


Windows PowerShell
Do you want to continue?
[Y] Yes [A] Yes to All [N] No [L] No to All [S] Suspend [?] Help (default is "N"): y
StatusCode   : 200
StatusDescription : OK
Content   : {"hostname":"DESKTOP-K8IJVKU","ledger_size":0,"node_id":1,"status":"alive"}
RawContent   : HTTP/1.1 200 OK
Connection: close
Content-Length: 76
Content-Type: application/json
Date: Sun, 07 Jun 2026 11:39:14 GMT
Server: Werkzeug/3.1.8 Python/3.13.13
{"hostname":"DESKTOP-K8IJVKU","led...
Forms   : {}
Headers   : {[Connection, close], [Content-Length, 76], [Content-Type, application/json], [Date, Sun, 07 Jun 2026 11:39:14 GMT]...}
Images   : {}
InputFields   : {}
Links   : {}
ParsedHtml   : mshtml.HTMLDocumentClass
RawContentLength : 76

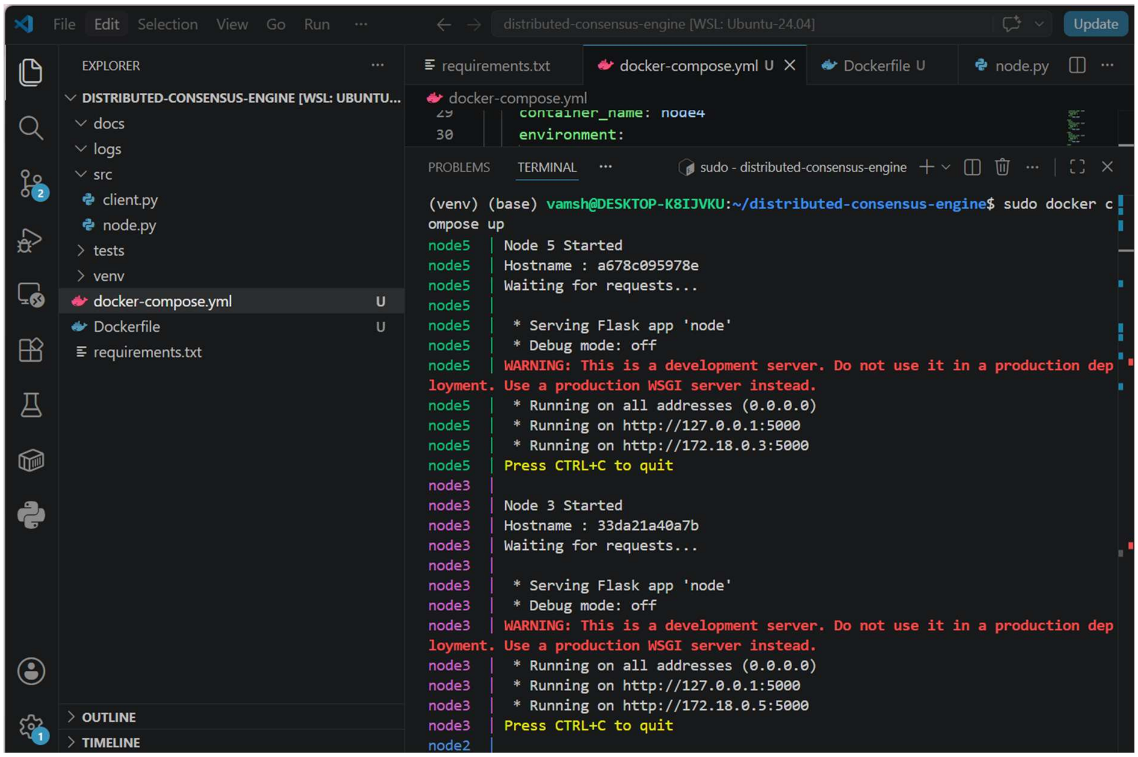

(base) vamsh@DESKTOP-K8IJVKU:/mnt/c/Users/Vamsh$ curl -X POST http://localhost:5000/transaction -H "Content-Type: application/json" -d '{"txn":"A pays B 100"}'
{"ledger_size":1,"node_id":1,"status":"committed","transaction":{"txn":"A pays B 100"}}

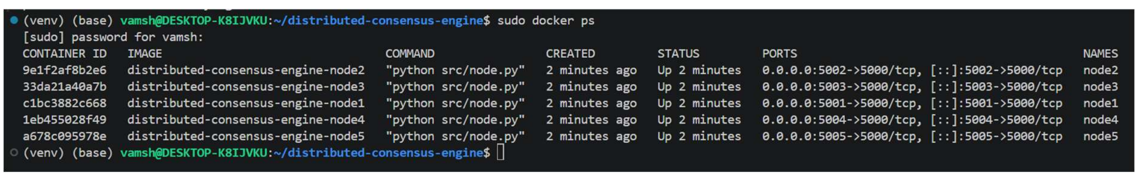


Build Docker Image

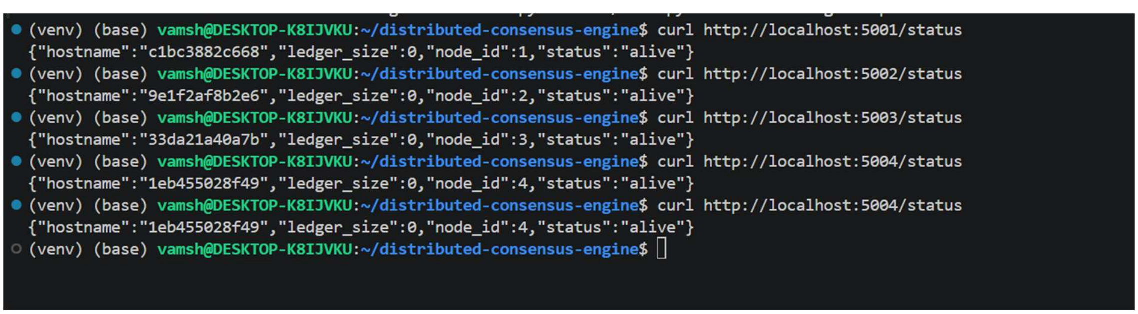


docker compose up and running, all nodes are started





This proves Docker networking works.



Send Transaction to Node1

```
(base) vamsh@DESKTOP-K8IJVKU:/mnt/c/Users/Vamsh$ curl -X POST http://localhost:5001/transaction \
-H "Content-Type: application/json" \
-d '{"txn":"A pays B 100"}'
{"ledger_size":1,"node_id":1,"status":"committed","transaction":{"txn":"A pays B 100"}}
(base) vamsh@DESKTOP-K8IJVKU:/mnt/c/Users/Vamsh$
```

Check Ledger

Pretty-print

{"ledger":[{"txn":"A pays B 100"}],"node_id":1}

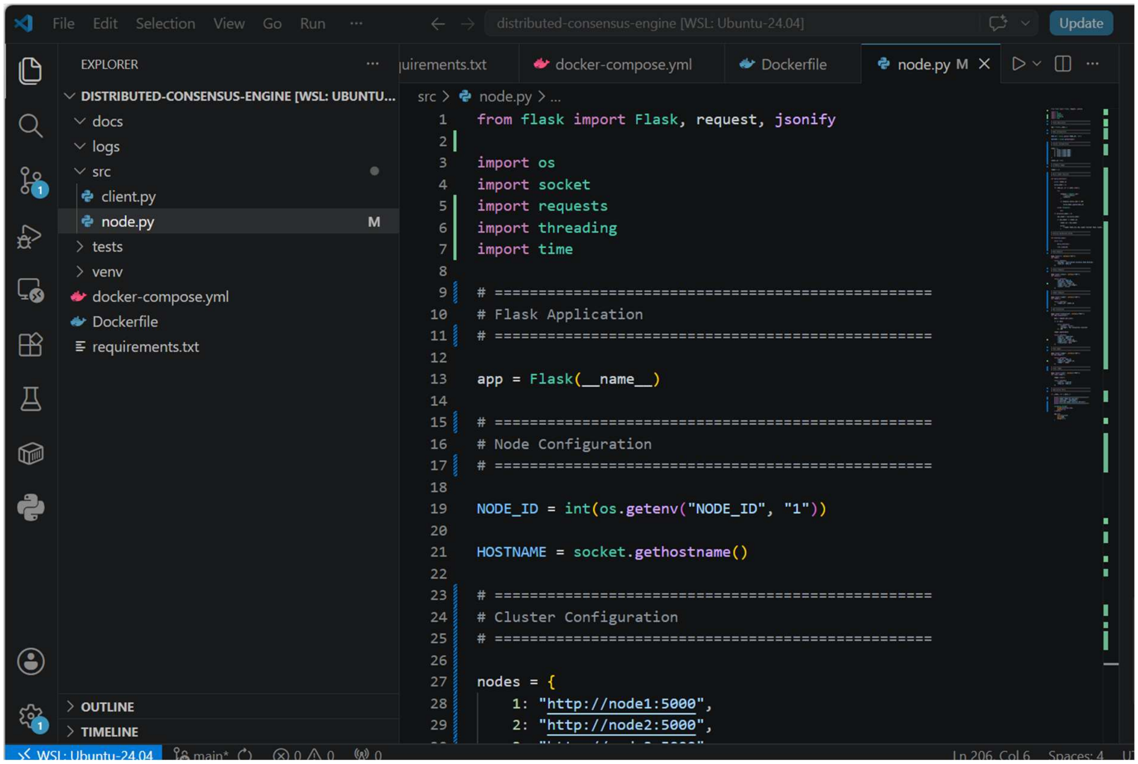
Verify Node Isolation

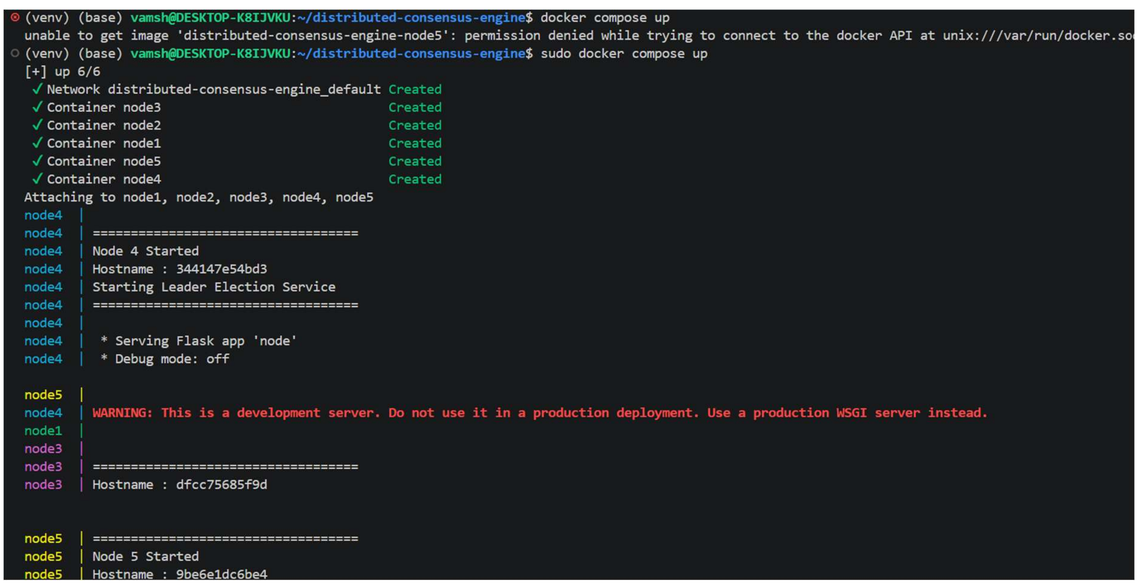
Pretty-print

{"ledger":[],"node_id":2}

```
PROBLEMS   TERMINAL   ...   bash - distributed-consensus-engine
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ sudo docker c
ompose up
Container node4 Stopping
Container node3 Stopping
Container node2 Stopping
Container node1 Stopped
node1 exited with code 137
Container node2 Stopped
node2 exited with code 137
Container node3 Stopped
node3 exited with code 137
Container node5 Stopped
node5 exited with code 137
Container node4 Stopped
node4 exited with code 137
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$
```

Leader Election (Bully Algorithm)





Check Leader

```
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5002/leader
{"leader_id":5}
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5001/leader
{"leader_id":5}
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5003/leader
{"leader_id":5}
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5004/leader
{"leader_id":5}
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5005/leader
{"leader_id":5}
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$
```

Simulate Leader Failure

```
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ sudo docker stop node5
[sudo] password for vamsh:
node5
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5001/leader
{"leader_id":4}
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$
```

Recover Leader

```
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ sudo docker start node5
node5
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5001/leader
{"leader_id":5}
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$
```

Heartbeats : Detect leader failure via heartbeats and elect a new leader automatically

```
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ sudo docker stop node5
[sudo] password for vamsh:
```

```
[Node 1] Leader timeout detected
[Node 1] New Leader Elected: Node 4
```

```
localhost:5001/leader
Pretty-print

{"leader_id":4}
```

```
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ sudo docker start node5
node5
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$
```

```
localhost:5001/leader
Pretty-print

{"leader_id":5}
```

Phase 12 : Paxos implementation

```
localhost:5001/leader
Pretty-print

{"leader_id":5}
```

```
(base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$  source /home/vamsh/distributed-consensus-engine/venv/bin/activate
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5001/leader
{"leader_id":5}
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl -X POST http://localhost:5005/propose_transaction \
-H "Content-Type: application/json" \
-d '{"txn":"A pays B 100"}'
{"leader":5,"status":"committed"}
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5005/ledger
{"leader_id":5,"ledger":[{"txn":"A pays B 100"}],"node_id":5}
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$
```

Phase 13: PBFT Cryptographic Signatures

Node1

↓

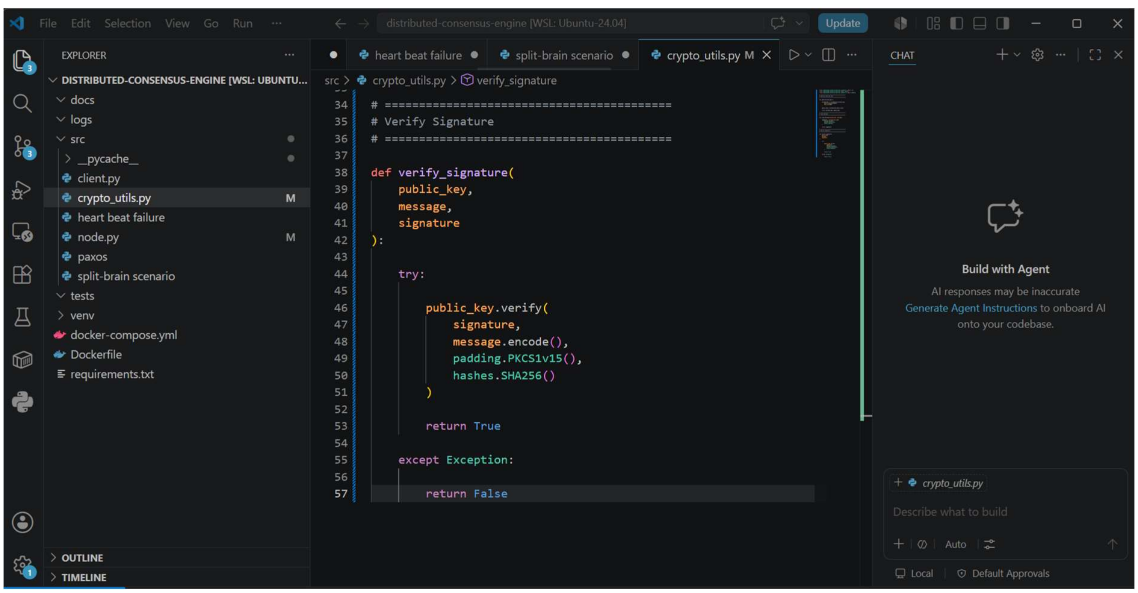
Digitally signs message

↓

Node2 verifies signature

↓

Accept only if valid




```
node3 exited with code 137
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5001/pbft_test
{"message":"TEST_PBFT","signature":"186ac14ee609561cc57b49a588d0854c96138630bbe2c4a356"}
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ tch    d Detach
```

```
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5001/pbft_test
{"message":"TEST_PBFT","signature":"186ac14ee609561cc57b49a588d0854c96138630bbe2c4a356"}
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5001/pbft_test
{"message":"TEST_PBFT","signature":"186ac14ee609561cc57b49a588d0854c96138630bbe2c4a356"}
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5002/pbft_test
{"message":"TEST_PBFT","signature":"2dae36d5f48fbe951684e7867d462afdb51c4467a16c6bae12"}
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5003/pbft_test
{"message":"TEST_PBFT","signature":"c51ee5b17f23c960c0ca64347fe2543dff1eb62d04e1fe56d0"}
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5004/pbft_test
{"message":"TEST_PBFT","signature":"752ff63c984e8aef817ba1dcebb432897a711e5b6517df01df"}
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5005/pbft_test
{"message":"TEST_PBFT","signature":"736652f814d73cb98f07ea2414fe3e424f6398257652bd0e42"}
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$
```

Client

|

Primary (Leader)

|

PRE-PREPARE

|

Replicas

|

PREPARE

|

Replicas

|

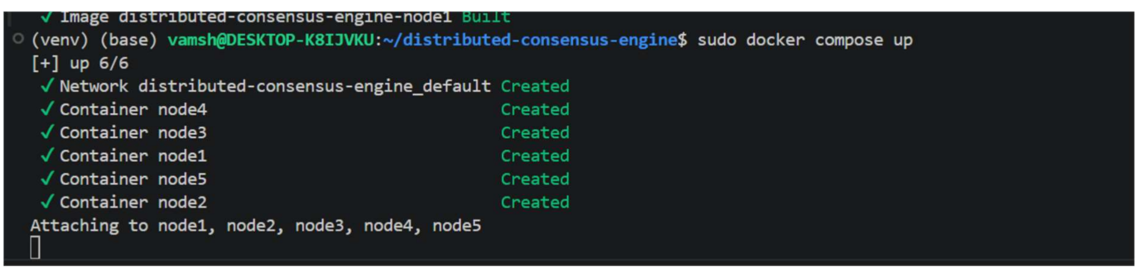
COMMIT

|

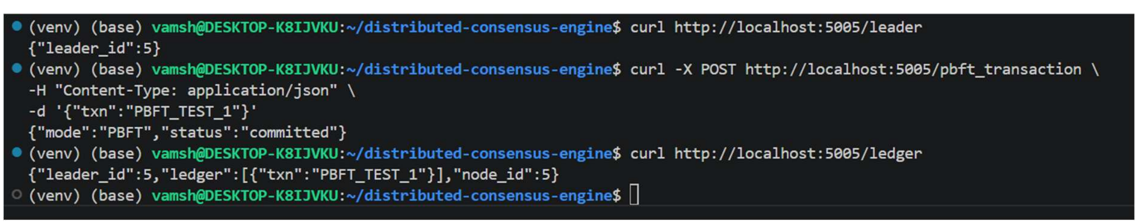
Consensus

|

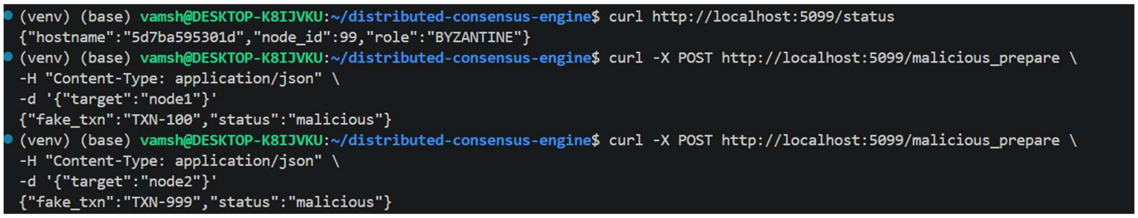
Ledger Commit



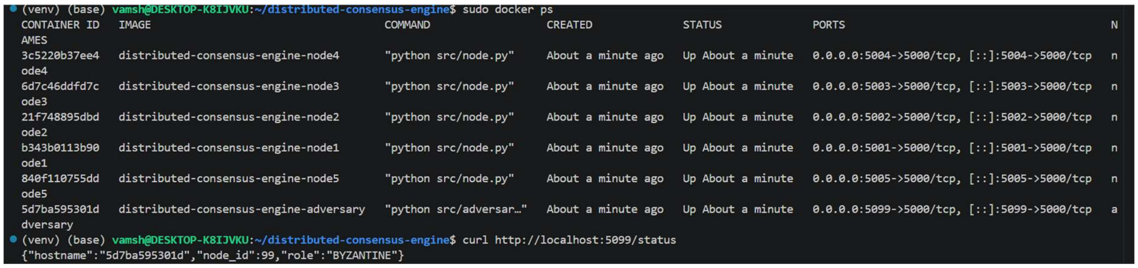
```
✓ Image distributed-consensus-engine-node1 Built
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ sudo docker compose up
[+] up 6/6
 ✓ Network distributed-consensus-engine_default Created
 ✓ Container node4                              Created
 ✓ Container node3                              Created
 ✓ Container node1                              Created
 ✓ Container node5                              Created
 ✓ Container node2                              Created
Attaching to node1, node2, node3, node4, node5
```



```
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5005/leader
{"leader_id":5}
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl -X POST http://localhost:5005/pbft_transaction \
-H "Content-Type: application/json" \
-d '{"txn":"PBFT_TEST_1"}'
{"mode":"PBFT","status":"committed"}
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5005/ledger
{"leader_id":5,"ledger":[{"txn":"PBFT_TEST_1"}],"node_id":5}
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$
```



```
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5099/status
{"hostname":"5d7ba595301d","node_id":99,"role":"BYZANTINE"}
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl -X POST http://localhost:5099/malicious_prepare \
-H "Content-Type: application/json" \
-d '{"target":"node1"}'
{"fake_txn":"TXN-100","status":"malicious"}
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl -X POST http://localhost:5099/malicious_prepare \
-H "Content-Type: application/json" \
-d '{"target":"node2"}'
{"fake_txn":"TXN-999","status":"malicious"}
```



```
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ sudo docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED              STATUS              PORTS                                         N
AMES
3c5220b37ee4   distributed-consensus-engine-node4   "python src/node.py"     About a minute ago   Up About a minute   0.0.0.0:5004->5000/tcp, [::]:5004->5000/tcp   n
ode4
6d7c46ddfd7c   distributed-consensus-engine-node3   "python src/node.py"     About a minute ago   Up About a minute   0.0.0.0:5003->5000/tcp, [::]:5003->5000/tcp   n
ode3
21f748895dbd   distributed-consensus-engine-node2   "python src/node.py"     About a minute ago   Up About a minute   0.0.0.0:5002->5000/tcp, [::]:5002->5000/tcp   n
ode2
b343b0113b90   distributed-consensus-engine-node1   "python src/node.py"     About a minute ago   Up About a minute   0.0.0.0:5001->5000/tcp, [::]:5001->5000/tcp   n
ode1
840f110755dd   distributed-consensus-engine-node5   "python src/node.py"     About a minute ago   Up About a minute   0.0.0.0:5005->5000/tcp, [::]:5005->5000/tcp   n
ode5
5d7ba595301d   distributed-consensus-engine-adversary   "python src/adversar…"   About a minute ago   Up About a minute   0.0.0.0:5099->5000/tcp, [::]:5099->5000/tcp   a
dversary
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ curl http://localhost:5099/status
{"hostname":"5d7ba595301d","node_id":99,"role":"BYZANTINE"}
```

chaos_test.sh

This below figures shows the successful completion of the chaos testing process. The distributed cluster remained operational after multiple failures, recoveries, and network partitions.

Observation: The testing workflow completed without causing total system failure.

Outcome: The implementation demonstrated resilience, recovery capability, and operational continuity under realistic distributed-system fault scenarios.

Start cluster

↓

Send transactions

↓

Kill leader

↓

[Verify re-election]

↓

Restart leader

↓

Partition network

↓

Recover network

↓

Continue transactions

```
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$ sudo bash tests/chaos_test.sh
PBFT / PAXOS CHAOS TEST STARTED
====================================

Current Containers
CONTAINER ID   IMAGE                                  COMMAND                  CREATED         STATUS         PORTS                                         NAMES
85802f040faf   distributed-consensus-engine-adversary "python src/adversar…"   8 minutes ago   Up 8 minutes   0.0.0.0:5099->5000/tcp, [::]:5099->5000/tcp   adversary
db0b0274ce1b   distributed-consensus-engine-node3     "python src/node.py"     8 minutes ago   Up 8 minutes   0.0.0.0:5003->5000/tcp, [::]:5003->5000/tcp   node3
3928d2f8da2f   distributed-consensus-engine-node5     "python src/node.py"     8 minutes ago   Up 8 minutes   0.0.0.0:5005->5000/tcp, [::]:5005->5000/tcp   node5
6df1a593a62d   distributed-consensus-engine-node4     "python src/node.py"     8 minutes ago   Up 8 minutes   0.0.0.0:5004->5000/tcp, [::]:5004->5000/tcp   node4
6d427d3640dd   distributed-consensus-engine-node1     "python src/node.py"     8 minutes ago   Up 8 minutes   0.0.0.0:5001->5000/tcp, [::]:5001->5000/tcp   node1
fafd2f6a41bb   distributed-consensus-engine-node2     "python src/node.py"     8 minutes ago   Up 8 minutes   0.0.0.0:5002->5000/tcp, [::]:5002->5000/tcp   node2

Waiting for cluster stabilization...

Checking Leader
{"leader_id":5}


====================================
TEST 1 : NORMAL TRANSACTIONS
====================================
{"leader":5,"status":"committed"}
{"leader":5,"status":"committed"}

Leader Ledger
{"leader_id":5,"ledger":[{"txn":"NORMAL_TXN_1"},{"txn":"NORMAL_TXN_2"}],"node_id":5}


====================================
TEST 2 : LEADER CRASH
====================================
```

```
====================================
TEST 2 : LEADER CRASH
====================================
node5

Leader stopped

New Leader
{"leader_id":4}


====================================
TEST 3 : TRANSACTION AFTER FAILURE
====================================
{"status":"failed"}
{"leader_id":4,"ledger":[],"node_id":4}


====================================
TEST 4 : LEADER RECOVERY
====================================
node5

Leader Status
{"leader_id":5}


====================================
TEST 5 : BYZANTINE NODE
====================================
{"hostname":"85802f040faf","node_id":99,"role":"BYZANTINE"}
```

```
======================================
TEST 5 : BYZANTINE NODE
======================================
{"hostname":"85802f040faf","node_id":99,"role":"BYZANTINE"}


{"fake_txn":"TXN-100","status":"malicious"}


{"fake_txn":"TXN-999","status":"malicious"}


======================================
TEST 6 : NETWORK PARTITION
======================================

Node3 isolated

Remaining Cluster
{"leader_id":5}


======================================
TEST 7 : RECOVER PARTITION
======================================

Node3 rejoined cluster
{"hostname":"db0b0274ce1b","leader_id":5,"ledger_size":0,"node_id":3,"status":"alive"}
```

```
======================================
TEST 8 : FINAL TRANSACTION
======================================
{"status":"failed"}
{"leader_id":5,"ledger":[],"node_id":5}


======================================
CHAOS TEST COMPLETED
======================================
(venv) (base) vamsh@DESKTOP-K8IJVKU:~/distributed-consensus-engine$
```